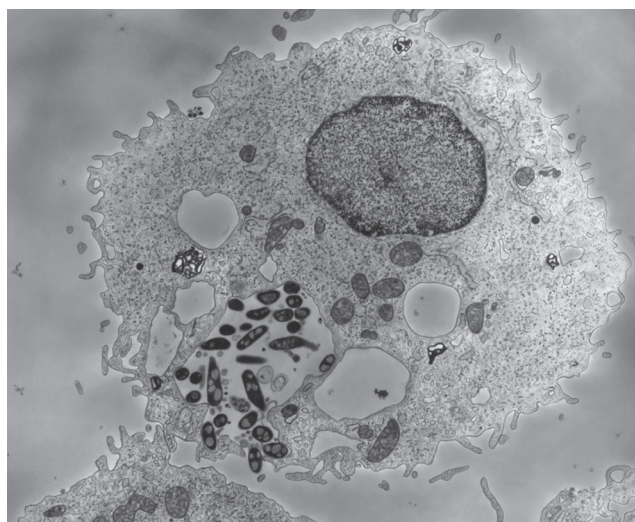


- 5 The vaccine used to control tuberculosis (TB) is known as Bacillus Calmette-Guérin (BCG). The vaccine contains live bacteria that have been selected so that they do not cause disease in humans.

Fig. 5.1 shows a macrophage that is in the process of engulfing the bacteria in the vaccine.



magnification $\times 4400$

Fig. 5.1

- (a) (i) Name the pathogen that causes TB.

..... [1]

- (ii) Describe how this pathogen is transmitted.

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..... [2]

- [5]

- Explain the role of memory cells in the body's defence against the TB pathogen.
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- [3]

(d) Suggest why vaccination with BCG has not yet eradicated TB.

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..... [3]

[Total: 14]